

14.10 The Exterior Algebra, and Clifford at Last

The exterior powers have so far been used one degree at a time, and the wedge of two of them has had nowhere to live: $\Lambda^2 V$ and $\Lambda^3 V$ are different spaces, and until now nothing said what $(\mathbf{u} \wedge \mathbf{w}) \wedge \mathbf{x}$ means as an operation. Stacking them fixes that, exactly as stacking the tensor powers produced the tensor algebra (Definition 14.6.2) and stacking the symmetric powers produced the symmetric algebra (Definition 14.7.3). The result is a 2^n -dimensional algebra, and it is the last object this chapter needs. With it we can finally pay the debt Chapter 13 recorded in writing: a Clifford algebra of (V, q) exists, and its dimension is 2^n .

Throughout, F is a field and V is a vector space over F with $\dim V = n$. From the subsection on decomposability on, $\text{char } F \neq 2$, because Definition 13.12.2 assumes it. We use the exterior powers of Definition 14.8.2 with their universal property for alternating maps, their bases of increasing wedges (Theorem 14.8.5), the test Theorem 14.8.7, and the reordering rule Lemma 14.8.1 (b) of the previous section.

14.10.1 Stacking the exterior powers

The pieces are already in place; only the multiplication is missing. The pattern is the one Proposition 14.7.4 used for symmetric powers: a product of a k -fold wedge and an l -fold wedge ought to be the $(k + l)$ -fold wedge of all the vectors, and the only thing to check is that this depends on the two factors and not on the way each was written.

Proposition 14.10.1. The Wedge of Two Exterior Powers

Let $k, l \geq 1$. There is a **unique** bilinear map

$$\Lambda^k V \times \Lambda^l V \rightarrow \Lambda^{k+l} V, \quad (\omega, \eta) \mapsto \omega \wedge \eta,$$

such that for all $\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_1, \dots, \mathbf{w}_l \in V$,

$$(\mathbf{u}_1 \wedge \dots \wedge \mathbf{u}_k) \wedge (\mathbf{w}_1 \wedge \dots \wedge \mathbf{w}_l) = \mathbf{u}_1 \wedge \dots \wedge \mathbf{u}_k \wedge \mathbf{w}_1 \wedge \dots \wedge \mathbf{w}_l.$$

Idea. Freeze the second factor's vectors and use the universal property once, in the first k slots; that

Definition 14.8.2 it induces a unique linear map

$$R(\mathbf{w}_1, \dots, \mathbf{w}_l): \Lambda^k V \rightarrow \Lambda^{k+l} V$$

with $R(\mathbf{w}_1, \dots, \mathbf{w}_l)(\mathbf{u}_1 \wedge \dots \wedge \mathbf{u}_k)$ equal to the displayed wedge.

Now let the $\mathbf{w}$'s vary. The assignment $(\mathbf{w}_1, \dots, \mathbf{w}_l) \mapsto R(\mathbf{w}_1, \dots, \mathbf{w}_l)$, with values in the vector space $\mathcal{L}(\Lambda^k V, \Lambda^{k+l} V)$, is l -linear and alternating: two linear maps on $\Lambda^k V$ are equal as soon as they agree on the wedges $\mathbf{u}_1 \wedge \dots \wedge \mathbf{u}_k$, which span $\Lambda^k V$ (Theorem 14.8.5), and on those the required identities are the linearity and the alternation of the wedge of $k + l$ vectors in its last l slots. So Definition 14.8.2 applies again and gives a unique linear map

$$\Lambda^l V \rightarrow \mathcal{L}(\Lambda^k V, \Lambda^{k+l} V), \quad \mathbf{w}_1 \wedge \dots \wedge \mathbf{w}_l \mapsto R(\mathbf{w}_1, \dots, \mathbf{w}_l).$$

Reading a linear map into a space of linear maps as a bilinear map gives the required product, with the stated values. It is unique because the wedges span both factors and a bilinear map is determined on a pair of spanning sets. This proves the proposition. ■

With the product in hand, the algebra assembles itself.

The exterior algebra is all the exterior powers at once, multiplied by wedging.

Definition 14.10.1. Exterior Algebra

Let V be a vector space over F , with $\Lambda^0 V = F$ and $\Lambda^1 V = V$ as in Definition 14.8.2. The **exterior algebra**, or **Grassmann algebra**, of V is the vector space

$$\Lambda V := \bigoplus_{k \geq 0} \Lambda^k V,$$

whose elements are the sequences $x = (x_0, x_1, \dots)$ with $x_k \in \Lambda^k V$ and only fin

- (b) The 2^n elements $\mathbf{e}_S$, one for each subset $S \subseteq [n]$, form a basis of ΛV . In particular $\dim_F \Lambda V = 2^n$.
- (c) ΛV is generated as an algebra by 1 and V .

Proof.

- (a) Bilinearity of the product holds componentwise, by [Proposition 14.10.1](#). For associativity, both $(x \wedge y) \wedge z$ and $x \wedge (y \wedge z)$ are trilinear in (x, y, z) , so it is enough to compare them on homogeneous elements, and then on the wedges of vectors, which span each $\Lambda^k V$ for $k \geq 1$ (Theorem 14.8.5) while $\Lambda^0 V$ is spanned by 1. On those both sides are the single wedge of all the listed vectors in order, by the defining identity of [Proposition 14.10.1](#) applied twice. The scalar 1 acts on each $\Lambda^k V$ as multiplication by 1, hence as the identity. The grading statement is built into the definition of the product.
- (b) By Theorem 14.8.5 the $\mathbf{e}_S$ with $|S| = k$ form a basis of $\Lambda^k V$, and $\Lambda^k V = \{\mathbf{0}\}$ for $k > n$. A basis of a direct sum is the union of bases of the summands, so the $\mathbf{e}_S$ over all $S \subseteq [n]$ form a basis of ΛV . Their number is $\sum_{k=0}^n \binom{n}{k} = 2^n$, by the binomial theorem applied to $(1+1)^n$.
- (c) Each $\mathbf{e}_S$ is a product of elements of V by the defining identity of the product, and the $\mathbf{e}_S$ span by (b).

■

The one structural fact that is not inherited from the tensor algebra is how the order of a product matters.

Theorem 14.10.3. Graded Commutativity

Let $\omega \in \Lambda^k V$ and $\eta \in \Lambda^l V$. Then

$$\omega \wedge \eta = (-1)^{kl} \eta \wedge \omega.$$

In particular $\omega \wedge \omega = \mathbf{0}$ whenever k is **odd**, over every field.

Idea. Count the swaps. To turn $\mathbf{w}_1 \wedge \cdots \wedge \mathbf{w}_l \wedge \mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_k$ into $\mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_k \wedge \mathbf{w$

A term with $S = T$ has a repeated vector, so $\mathbf{e}_S \wedge \mathbf{e}_S = \mathbf{0}$. A pair $S \neq T$ occurs twice, and the two occurrences are $c_{SCT} \mathbf{e}_S \wedge \mathbf{e}_T$ and $c_{TCS} \mathbf{e}_T \wedge \mathbf{e}_S = -c_{SCT} \mathbf{e}_S \wedge \mathbf{e}_T$, since $(-1)^{kl} = (-1)^{k^2} = -1$ for odd k . So the two cancel, and nothing divides by 2 anywhere. ■

Warning. Graded commutative is not commutative. In ΛF^2 we have $\mathbf{e}_1 \wedge \mathbf{e}_2 = -\mathbf{e}_2 \wedge \mathbf{e}_1$, and these are different elements unless $\text{char } F = 2$. What is true in all degrees is only the signed rule. And the sign is $+1$ as soon as one of the degrees is even, so an element of $\Lambda^2 V$ commutes with everything, while two elements of $\Lambda^1 V = V$ anticommute.

Warning. $\omega \wedge \omega$ need not be zero. The identity $\mathbf{v} \wedge \mathbf{v} = \mathbf{0}$ is a statement about degree 1. For $\omega = \mathbf{e}_1 \wedge \mathbf{e}_2 + \mathbf{e}_3 \wedge \mathbf{e}_4$ in ΛF^4 , a direct expansion gives $\omega \wedge \omega = 2 \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_4$, which is non-zero whenever $\text{char } F \neq 2$. Squaring to zero is the privilege of odd degree.

Example. The exterior algebra of a three-dimensional space

Describe ΛF^3 completely, and compute $(\mathbf{e}_1 + \mathbf{e}_2 \wedge \mathbf{e}_3) \wedge (\mathbf{e}_2 + \mathbf{e}_1 \wedge \mathbf{e}_3)$.

Solution. By Theorem 14.10.2 it has dimension $2^3 = 8$, with basis

$$1; \quad \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3; \quad \mathbf{e}_{12}, \mathbf{e}_{13}, \mathbf{e}_{23}; \quad \mathbf{e}_{123},$$

writing $\mathbf{e}_{ij}$ for $\mathbf{e}_i \wedge \mathbf{e}_j$ and $\mathbf{e}_{123}$ for $\mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3$. The degrees have dimensions 1, 3, 3, 1. Every product of two elements of degree ≥ 2 lands in degree $\geq$

14.10.2 Decomposable elements

Section 2 warned that not every element of $V \otimes W$ is a simple tensor $\mathbf{v} \otimes \mathbf{w}$, and proved it with $\mathbf{e}_1 \otimes \mathbf{e}_1 + \mathbf{e}_2 \otimes \mathbf{e}_2$ (Example (A tensor that is not simple)). The same thing happens one floor up, and for the same reason: the simple objects form a low-dimensional cone inside a large space, and a sum of two of them usually escapes it.

Definition 14.10.2. Decomposable Element

Let $k \geq 1$. An element $\omega \in \Lambda^k V$ is **decomposable**, or a **k -blade**, if there exist $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$ with

$$\omega = \mathbf{v}_1 \wedge \mathbf{v}_2 \wedge \cdots \wedge \mathbf{v}_k.$$

Examples. Every element of $\Lambda^1 V = V$ is decomposable, with $k = 1$; so is $\mathbf{0}$, being $\mathbf{v} \wedge \mathbf{v} \wedge \mathbf{v}_3 \wedge \cdots \wedge \mathbf{v}_k$; and so is every element of the line $\Lambda^n V$, since $c\mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_n = (c\mathbf{e}_1) \wedge \mathbf{e}_2 \wedge \cdots \wedge \mathbf{e}_n$.

Why the word. A decomposable $\omega \neq \mathbf{0}$ carries a k -dimensional subspace: its factors are independent by Theorem 14.8.7, and they span one. Non-decomposable elements are exactly those that come from no single subspace, which is why the failure below is a phenomenon and not a notational accident.

Theorem 14.10.4. Not Every 2-Vector Is Decomposable

Let $\dim V = n \geq 4$, let $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ be a basis of V , and set

$$\omega = \mathbf{e}_1 \wedge \mathbf{e}_2 + \mathbf{e}_3 \wedge \mathbf{e}_4 \in \Lambda^2 V.$$

Then ω is **not** decomposable.

Idea. A decomposable 2-vector $\mathbf{u} \wedge \math$

of $\Lambda^3 V$, with coefficients x_3, x_4, x_1, x_2 and x_i, x_i respectively, up to signs. Since these basis vectors are distinct and the $\mathbf{e}_S$ are independent (Theorem 14.8.5), $\mathbf{v} \wedge \omega = \mathbf{0}$ forces $x_1 = x_2 = \cdots = x_n = 0$, that is, $\mathbf{v} = \mathbf{0}$. This contradicts $\mathbf{u} \neq \mathbf{0}$ with $\mathbf{u} \wedge \omega = \mathbf{0}$. Hence ω is not decomposable. ■

Remark. When $\text{char } F \neq 2$ there is a one-line version: $\omega \wedge \omega = 2 \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_4 \neq \mathbf{0}$, whereas $(\mathbf{u} \wedge \mathbf{w}) \wedge (\mathbf{u} \wedge \mathbf{w}) = \mathbf{0}$ always. The proof above was written to avoid that division, because the test fails over $\mathbb{F}_2$ — see the warning next.

Warning. **The square test is blind in characteristic 2.** Over $\mathbb{F}_2$ every $\omega \in \Lambda^2 V$ satisfies $\omega \wedge \omega = \mathbf{0}$: in the expansion $\sum_{S,T} c_S c_T \mathbf{e}_S \wedge \mathbf{e}_T$ the diagonal terms vanish and each off-diagonal pair contributes $2c_S c_T \mathbf{e}_S \wedge \mathbf{e}_T = \mathbf{0}$. So over $\mathbb{F}_2$ the vanishing of $\omega \wedge \omega$ says nothing at all, while $\mathbf{e}_1 \wedge \mathbf{e}_2 + \mathbf{e}_3 \wedge \mathbf{e}_4$ is still not decomposable, by the proof above. A criterion and a phenomenon are different things.

Quick check. Why does Theorem 14.10.4 need $n \geq 4$? Show that every element of $\Lambda^2 V$ is decomposable when $\dim V = 3$.

Solution. With $n \leq 3$ there is no room to write $\mathbf{e}_1 \wedge \mathbf{e}_2 + \mathbf{e}_3 \wedge \mathbf{e}_4$ at all. For $n = 3$, take $\omega = a \mathbf{e}_{12} + b \mathbf{e}_{13} + c \mathbf{e}_{23}$. If $a = b = c = 0$ then $\omega = \mathbf{0}$, which is decomposable. If $a \neq 0$, then

$$\left(\mathbf{e}_1 - \frac{c}{a} \mathbf{e}_3 \right) \wedge (a \mathbf{e}_2 + b \mathbf{e}_3) = a \mathbf{e}_{12} + b \mathbf{e}_{13} + c \mathbf{e}_{23} = \omega,$$

Let A be an F -algebra (Definition 13.12.1). A **two-sided ideal** of A is a subspace $\mathcal{I} \subseteq A$ such that $yx \in \mathcal{I}$ and $xy \in \mathcal{I}$ for **every** $x \in \mathcal{I}$ and **every** $y \in A$. For a subset $S \subseteq A$, the **ideal generated by S** is

$$\langle S \rangle := \text{span}\{ ysz : y, z \in A, s \in S \}.$$

Examples. In $F[x]$ the ideals are the sets $\langle p \rangle$ of Definition 5.2.1; in any A , both $\{0\}$ and A are ideals. **Non-example by minimal change.** The diagonal matrices in $M_2(F)$ form a subspace closed under multiplication, but not an ideal: it contains $\mathbf{E}_{11}$, while $\mathbf{E}_{21}\mathbf{E}_{11} = \mathbf{E}_{21}$ is not diagonal. Being closed among one's own elements is not the condition; absorbing products with the **whole** algebra is.

The span above is a subspace by construction, and it is a two-sided ideal because $a(ysz) = (ay)sz$ and $(ysz)a = ys(za)$ are again of the listed form. It contains S , taking $y = z = 1_A$, and every two-sided ideal containing S contains each ysz ; so $\langle S \rangle$ is the smallest two-sided ideal containing S , which is what “generated” means.

Lemma 14.10.5. Quotient of an Algebra by an Ideal

Let A be an F -algebra and $\mathcal{I}$ a two-sided ideal of A .

- (a) The quotient space $A/\mathcal{I}$ carries exactly one multiplication making the quotient map $\pi: A \rightarrow A/\mathcal{I}$ an algebra homomorphism, namely $(x + \mathcal{I})(y + \mathcal{I}) = xy + \mathcal{I}$; with it $A/\mathcal{I}$ is an F -algebra with identity $1_A + \mathcal{I}$, and $\ker \pi = \mathcal{I}$.
- (b) If $\varphi: A \rightarrow B$ is an algebra homomorphism with $\mathcal{I} \subseteq \ker \varphi$, then there is exactly one algebra homomorphism $\bar{\varphi}: A/\mathcal{I} \rightarrow B$ with $\bar{\varphi} \circ \pi = \varphi$.
- (c) The kernel of any algebra homomorphism is a two-sided ideal.

Proof.

14.10.4 The Clifford algebra, constructed

Chapter 13 §12 defined a Clifford algebra of (V, q) as a pair (C, ι) with $\iota: V \rightarrow C$ linear satisfying

- (C1) $\iota(\mathbf{v})^2 = q(\mathbf{v})1_C$ for every $\mathbf{v} \in V$, and
- (C2) the universal property: every linear $f: V \rightarrow A$ into an F -algebra with $f(\mathbf{v})^2 = q(\mathbf{v})1_A$ factors as $f = \tilde{f} \circ \iota$ for a unique algebra homomorphism $\tilde{f}$,

and then said in as many words that it could not produce one: “The definition says what a Clifford algebra would be; it does not produce one.” What it needed was an algebra of formal strings of vectors, which is now Definition 14.6.2, and a way to impose the relation $\mathbf{v} \otimes \mathbf{v} = q(\mathbf{v})1$, which is now Lemma 14.10.5.

Theorem 14.10.6. Existence of the Clifford Algebra

Let F be a field with $\text{char } F \neq 2$, let V be a finite-dimensional F -vector space, let β be a symmetric bilinear form on V and $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$. Let

$$\mathcal{I}_q := \langle \mathbf{v} \otimes \mathbf{v} - q(\mathbf{v})1 : \mathbf{v} \in V \rangle \subseteq T(V)$$

be the two-sided ideal generated by these elements, put $\text{Cl}(q) := T(V)/\mathcal{I}_q$, and let $\iota: V \rightarrow \text{Cl}(q)$ be the restriction to $V \subseteq T(V)$ of the quotient map π . Then $(\text{Cl}(q), \iota)$ is a Clifford algebra of (V, q) in the sense of Definition 13.12.2.

In particular a Clifford algebra exists for every (V, q) , and by Proposition 13.12.1 any two are isomorphic by a unique isomorphism compatible with the structure maps.

Idea. The tensor algebra is the free associative algebra on V : a linear map out of V extends uniquely to it (Theorem 14.6.6), with no relations in the way. Dividing by the smallest ideal containing $\mathbf{v} \otimes \mathbf{v} - q(\mathbf{v})$

For the uniqueness clause of (C2), let $g: \text{Cl}(q) \rightarrow A$ be any algebra homomorphism with $g \circ \iota = f$. Then $g \circ \pi$ is an algebra homomorphism $T(V) \rightarrow A$ whose restriction to V is $g\iota = f$, so $g\pi = \bar{f}$ by the uniqueness clause of Theorem 14.6.6. Hence $g\pi = \bar{f}\pi$, and π is surjective, so $g = \bar{f}$. This proves the theorem. ■

Remark. Neither $\text{char } F \neq 2$ nor finite dimension was used in this proof; the construction and both checks work for any V and any quadratic form. The hypotheses are carried because Definition 13.12.2 states them, and because the dimension count below needs an orthogonal basis.

14.10.5 The dimension, at last

Chapter 13 proved half of the count. Its Proposition 13.12.3 showed that for an orthogonal basis $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ of β the 2^n **sorted products**

$$\iota(\mathbf{e}_{i_1})\iota(\mathbf{e}_{i_2}) \cdots \iota(\mathbf{e}_{i_k}), \quad i_1 < i_2 < \cdots < i_k,$$

span C , so that $\dim_F C \leq 2^n$. The other half — their independence — is what this section owes. The idea is to make ΛV , which has exactly the right dimension, into a place where the vectors of V act with square $q(\mathbf{v})$; the universal property then sends C into the operators on ΛV , and evaluating at 1 turns the sorted products into distinct basis vectors.

Two families of operators on ΛV do the work: wedging a basis vector on, and deleting it again.

Lemma 14.10.7. Creation and Deletion Operators

Let $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ be a basis of V , and for $S \subseteq [n]$ let $\mathbf{e}_S$ be as in Theorem 14.10.2. For $i \in [n]$ write $\sigma(i, S) = |\{j \in S : j < i\}|$. Let $E_i \in \mathcal{L}(\Lambda V)$ be left multiplication by $\mathbf{e}_i$, that is, $E_i(\omega) = \mathbf{e}_i \wedge \omega$, and let $D_i \in \mathcal{L}(\Lambda V)$ be the unique linear map with

Proof. First record what E_i does on the basis. If $i \in S$ then $\mathbf{e}_i \wedge \mathbf{e}_S$ repeats $\mathbf{e}_i$ and is $\mathbf{0}$. If $i \notin S$, then sorting $\mathbf{e}_i \wedge \mathbf{e}_S$ moves $\mathbf{e}_i$ past the $\sigma(i, S)$ members of S smaller than i , so by Lemma 14.8.1 (b)

$$E_i(\mathbf{e}_S) = (-1)^{\sigma(i, S)} \mathbf{e}_{S \cup \{i\}}.$$

First identity. For every $\omega \in \Lambda V$, $E_i E_j(\omega) = \mathbf{e}_i \wedge \mathbf{e}_j \wedge \omega$ and $E_j E_i(\omega) = \mathbf{e}_j \wedge \mathbf{e}_i \wedge \omega$, and $\mathbf{e}_j \wedge \mathbf{e}_i = -\mathbf{e}_i \wedge \mathbf{e}_j$ by Theorem 14.10.3. Adding gives 0. (For $i = j$ each term is separately 0.)

Third identity, case $i = j$. If $i \notin S$, then $D_i(\mathbf{e}_S) = \mathbf{0}$ and $E_i(\mathbf{e}_S) = (-1)^{\sigma(i, S)} \mathbf{e}_{S \cup \{i\}}$; since $\sigma(i, S \cup \{i\}) = \sigma(i, S)$, applying D_i returns $(-1)^{2\sigma(i, S)} \mathbf{e}_S = \mathbf{e}_S$. If $i \in S$, then $E_i(\mathbf{e}_S) = \mathbf{0}$ and $D_i(\mathbf{e}_S) = (-1)^{\sigma(i, S)} \mathbf{e}_{S \setminus \{i\}}$; since $\sigma(i, S \setminus \{i\}) = \sigma(i, S)$, applying E_i returns $\mathbf{e}_S$. Either way the sum is $\mathbf{e}_S$.

Third identity, case $i \neq j$. If $j \in S$ then $E_j(\mathbf{e}_S) = \mathbf{0}$, and $D_i(\mathbf{e}_S)$ is either $\mathbf{0}$ or a multiple of $\mathbf{e}_{S \setminus \{i\}}$, which still contains j , so E_j kills it; both terms vanish. If $j \notin S$ and $i \notin S$ then $D_i(\mathbf{e}_S) = \mathbf{0}$ and $D_i(\mathbf{e}_{S \cup \{j\}}) = \mathbf{0}$ as well, since $i \neq j$; again both terms vanish. So let $j \notin S$ and $i \in S$. Then

$$\begin{aligned} D_i E_j(\mathbf{e}_S) &= (-1)^{\sigma(j, S) + \sigma(i, S \cup \{j\})} \mathbf{e}_{(S \cup \{j\}) \setminus \{i\}}, \\ E_j D_i(\mathbf{e}_S) &= (-1)^{\sigma(i, S)$$

Proof. An orthogonal basis exists by Theorem 13.4.2, since $\text{char } F \neq 2$. Write $d_i = q(\mathbf{e}_i)$ and $c_i = E_i + d_i D_i \in \mathcal{L}(\Lambda V)$, with E_i, D_i as in Lemma 14.10.7. By Proposition 13.12.3 the sorted products span C and $\dim_F C \leq 2^n$, so only independence is left.

Step 1: the c_i satisfy the Clifford relations. Expanding the product and collecting the four brackets,

$$\begin{aligned} c_i c_j + c_j c_i &= (E_i E_j + E_j E_i) + d_j (E_i D_j + D_j E_i) \\ &\quad + d_i (D_i E_j + E_j D_i) + d_i d_j (D_i D_j + D_j D_i). \end{aligned}$$

By Lemma 14.10.7 the first and last brackets are 0 and each middle bracket is $\delta_{ij} \text{id}$. So $c_i c_j = -c_j c_i$ for $i \neq j$. Setting $i = j$ in the display would give $2c_i^2$, so expand c_i^2 directly instead; with $E_i^2 = D_i^2 = 0$, which were checked separately in the lemma,

$$c_i^2 = E_i^2 + d_i (E_i D_i + D_i E_i) + d_i^2 D_i^2 = d_i \text{id},$$

with no division by 2 anywhere. Now $\mathcal{L}(\Lambda V)$ is an F -algebra under composition (Definition 13.12.1). Let $f: V \rightarrow \mathcal{L}(\Lambda V)$ be the linear map with $f(\mathbf{e}_i) = c_i$, which exists and is unique by Theorem 3.1.6. For $\mathbf{v} = \sum_i x_i \mathbf{e}_i$,

$$f(\mathbf{v})^2 = \sum_i x_i^2 c_i^2 + \sum_{i < j} x_i x_j (c_i c_j + c_j c_i) = \left(\sum_i d_i x_i^2 \right) \text{id},$$

the cross terms vanishing because $i \neq j$ there. Since the basis is orthogonal, $\beta(\mathbf{e}_i, \mathbf{e}_j) = 0$ for $i \neq j$, so $q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v}) = \sum_i d_i x_i^2$. Hence $f(\mathbf{v})^2 = q(\mathbf{v}) \text{id}$, which is (C1) for f .

Step 2: the induced homomorphism. By the universal property (C2) of Definition 13.12.2 there is an algebra homomorphism $\Psi: C \rightarrow \mathcal{L}(\$

Since 1_C and $\iota(\mathbf{e}_1), \dots, \iota(\mathbf{e}_n)$ are among the basis elements just found, the list $(\iota(\mathbf{e}_1), \dots, \iota(\mathbf{e}_n))$ is independent, so ι carries a basis of V to an independent list and is injective. Finally Φ carries the basis of C bijectively onto the basis $(\mathbf{e}_S)$ of ΛV , so Φ is a vector space isomorphism. This proves the theorem. ■

The debt, and how it was paid. Chapter 13 §12 stated Theorem 13.12.4 and then wrote, under the heading “What is proved here, and what is not”, that the inequality $\dim_F C \leq 2^n$ was proved in full, while “the reverse inequality — that the 2^n sorted products are linearly independent — is **not proved in this section**”, and that the two halves would have to be established together by exhibiting one algebra of dimension exactly 2^n satisfying (C1) and (C2). Both halves are now supplied. The missing one was independence, and Theorem 14.10.8 supplies it; the prior question, whether any Clifford algebra exists at all, is Theorem 14.10.6. Chapter 13 read the statement for $n \geq 3$ “as quoted from Chapter 14 and not as proved here”; it may now be read as proved.

The prediction was right in substance and slightly off in shape. The algebra of dimension 2^n that settled the matter is the exterior algebra, but it is not the Clifford algebra: it is only a space that $\text{Cl}(q)$ acts on, and independence came from the action rather than from an identification. Chapter 13 also guessed that “ $\text{Cl}(q)$ is assembled from the exterior powers $\Lambda^k V$, one for each k ”, and Φ is that assembly, matching the sorted product on S with the wedge $\mathbf{e}_S$.

Warning. $\text{Cl}(q)$ and ΛV are the same size, not the same algebra. The map Φ above is an isomorphism of **vector spaces** only. It is not multiplicative, and no other map is either, unless $q = 0$: in ΛV the square of a vector is $\mathbf{0}$, while in $\text{Cl}(q)$ it is $q(\mathbf{v})1$. Writing an element of $\text{Cl}(q)$ as a sum of wedges is a choice of coordinates, not a description of the multiplication.

That warning deserves a proof rather than a slogan.

Proposition 14.10.9. Comparing the Two Algebras

Let $\text{char } F \neq 2$ and $\dim V = n \geq 1$, and let q be the quadratic form of a symmetric bilinear form on V .

- (a) If q

there is an algebra homomorphism $\psi: \text{Cl}(0) \rightarrow \Lambda V$ with $\psi\iota = j$. Its image is a subalgebra containing 1 and V , hence all of ΛV by [Theorem 14.10.2 \(c\)](#); so ψ is surjective. Both spaces have dimension 2^n , by [Theorem 14.10.8](#) and [Theorem 14.10.2 \(b\)](#), so ψ is bijective and is an algebra isomorphism.

(b) *Claim.* If $x \in \Lambda V$ and $x \wedge x = a \cdot 1$ with $a \in F$, $a \neq 0$, then $x \in F \cdot 1$.

Proof of claim. Throughout, y^t means the t -fold wedge product of y with itself. Write $x = x_0 + m$ with $x_0 \in \Lambda^0 V = F$ and m the sum of the components of positive degree. Then

$$x \wedge x = x_0^2 + 2x_0m + m \wedge m,$$

and the only degree-0 term is x_0^2 , so $x_0^2 = a \neq 0$ and hence $x_0 \neq 0$. Canceling that term leaves

$$m \wedge (2x_0 \cdot 1 + m) = \mathbf{0}.$$

Every product of $n+1$ elements of positive degree lies in $\bigoplus_{k>n} \Lambda^k V = \{\mathbf{0}\}$, so $m^{n+1} = \mathbf{0}$. Put $c = 2x_0$, which is non-zero because $\text{char } F \neq 2$ and $x_0 \neq 0$, and $u = -c^{-1}m$, so that $u^{n+1} = \mathbf{0}$ and $c \cdot 1 + m = c(1 - u)$. Then

$$(1 - u) \wedge (1 + u + u^2 + \cdots + u^n) = 1 - u^{n+1} = 1,$$

and likewise in the other order, so $c \cdot 1 + m$ is invertible. Multiplying $m \wedge (c \cdot 1 + m) = \mathbf{0}$ by its inverse gives $m = \mathbf{0}$, so $x = x_0 \in F \cdot 1$, proving the claim.

Now suppose $q \neq 0$ and let $\varphi: \text{Cl}(q) \rightarrow \Lambda V$ be an algebra isomorphism. Choose $\mathbf{v} \$

$\bar{f}\psi^{-1}: A \rightarrow B$ is an algebra homomorphism with $\bar{f}\psi^{-1}\iota_A = \bar{f}\iota = f$, where $\bar{f}: C \rightarrow B$ comes from (C2) for C ; and if $g: A \rightarrow B$ is another one, then $g\psi: C \rightarrow B$ satisfies $g\psi\iota = g\iota_A = f$, so $g\psi = \bar{f}$ by uniqueness for C , whence $g = \bar{f}\psi^{-1}$. ■

Example. The Clifford algebra of the Euclidean plane

Let $V = \mathbb{R}^2$ with $q(\mathbf{x}) = x_1^2 + x_2^2$, the form of signature $(2, 0)$. Show that $\text{Cl}_{2,0}(\mathbb{R}) \cong M_2(\mathbb{R})$.

Solution. Put

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

in $M_2(\mathbb{R})$, and let $\iota(\mathbf{x}) = x_1\mathbf{A} + x_2\mathbf{B}$, a linear map $\mathbb{R}^2 \rightarrow M_2(\mathbb{R})$. Multiplying out, $\mathbf{A}^2 = \mathbf{B}^2 = \mathbf{I}_2$ and $\mathbf{AB} = -\mathbf{BA}$, the products being $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and its negative. Hence

$$\iota(\mathbf{x})^2 = x_1^2\mathbf{A}^2 + x_1x_2(\mathbf{AB} + \mathbf{BA}) + x_2^2\mathbf{B}^2 = q(\mathbf{x})\mathbf{I}_2,$$

which is (C1). Next, $\dim_{\mathbb{R}} M_2(\mathbb{R}) = 4 = 2^2$, and the four matrices $\mathbf{I}_2, \mathbf{A}, \mathbf{B}, \mathbf{AB}$ are linearly independent: the first two span the diagonal matrices and the last two span the off-diagonal ones, and the diagonal and off-diagonal matrices meet only in 0. So they are a basis, and $M_2(\mathbb{R})$ is generated by $\mathbf{I}_2$ and $\iota(\mathbb{R}^2)$. By [Corollary 14.10.10](#), $(M_2(\mathbb{R}), \iota)$ is a Clifford algebra of $(\mathbb{R}^2, q)$, so $\text{Cl}_{2,0}(\mathbb{R}) \cong M_2(\mathbb{R})$.

Compare Chapter 13's $\text{Cl$

Exercise. A1

- (a) Define the exterior algebra ΛV and state its dimension when $\dim V = n$.
- (b) State the graded commutativity rule, and say what it gives when both degrees are odd.
- (c) Define what it means for $\omega \in \Lambda^k V$ to be decomposable, and give one decomposable and one non-decomposable element of $\Lambda^2 F^4$.
- (d) Which half of Theorem 13.12.4 was left unproved in Chapter 13, and which result of this section supplies it?
- (e) True or false: $\text{Cl}(q)$ and ΛV are isomorphic algebras. Justify your answer.

Solution.

- (a) $\Lambda V = \bigoplus_{k \geq 0} \Lambda^k V$ with the product of [Proposition 14.10.1](#) extended by bilinearity ([Definition 14.10.1](#)). Its dimension is $\sum_k \binom{n}{k} = 2^n$ ([Theorem 14.10.2](#)).
- (b) $\omega \wedge \eta = (-1)^{kl} \eta \wedge \omega$ for $\omega \in \Lambda^k V$, $\eta \in \Lambda^l V$ ([Theorem 14.10.3](#)). Both odd gives $\omega \wedge \eta = -\eta \wedge \omega$: odd-degree elements anticommute.
- (c) ω is decomposable if $\omega = \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k$ for some $\mathbf{v}_i \in V$ ([Definition 14.10.2](#)). Decomposable: $\mathbf{e}_1 \wedge \mathbf{e}_2$. Non-decomposable: $\mathbf{e}_1 \wedge \mathbf{e}_2 + \mathbf{e}_3 \wedge \mathbf{e}_4$ ([Theorem 14.10.4](#)).
- (d) The independence of the 2^n sorted products, that is, the inequality $\dim_F C \geq 2^n$. It is supplied by [Theorem 14.10.8](#), which needs [Theorem 14.10.6](#) first so that the object exists.
- (e) False in general. They are isomorphic as vector spaces, and as algebras exactly when $q = 0$ ([Proposition 14.10.9](#)).

14.10.6.2 B. Practice**Exercise. B1: Multiplying in the exterior algebra**

In ΛF^4 with $F = \mathbb{Q}$, put $x = \mathbf{e}_1 + \mathbf{e}_2 \wedge \mathbf{e}_3$ and $y = \mathbf{e}_2 + \mathbf{e}_1 \wedge \mathbf{e}_4$. Compute the following, expressed in the basis of [Theorem 14.10.2](#).

the $(1, 2)$ and $(2, 2)$ parts do not.

- (c) $x \wedge x = \mathbf{e}_1 \wedge \mathbf{e}_1 + \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 + \mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_1 + \mathbf{0} = 2\mathbf{e}_{123}$, since $\mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_1 = \mathbf{e}_{123}$. An inhomogeneous element need not square to zero.

Exercise. B2: Decomposable or not

In $\Lambda^2 \mathbb{Q}^4$, writing $\mathbf{e}_{ij} = \mathbf{e}_i \wedge \mathbf{e}_j$, determine whether each of the following is decomposable. Justify your answer.

$$\omega_1 = \mathbf{e}_{12} + \mathbf{e}_{13} + 3\mathbf{e}_{14} - 2\mathbf{e}_{23} - \mathbf{e}_{24} + 5\mathbf{e}_{34}, \quad \omega_2 = \mathbf{e}_{12} + \mathbf{e}_{34} + \mathbf{e}_{14}.$$

Solution. For $\omega = \sum_{i < j} c_{ij} \mathbf{e}_{ij}$ in $\Lambda^2 \mathbb{Q}^4$, expanding gives

$$\omega \wedge \omega = 2(c_{12}c_{34} - c_{13}c_{24} + c_{14}c_{23}) \mathbf{e}_{1234},$$

since the only pairs of index sets with empty intersection are $\{12\}, \{34\}$ and $\{13\}, \{24\}$ and $\{14\}, \{23\}$, each occurring in two orders, and $\mathbf{e}_{13} \wedge \mathbf{e}_{24} = -\mathbf{e}_{1234}$ while the other two products are $+\mathbf{e}_{1234}$.

For ω_1 : $1 \cdot 5 - 1 \cdot (-1) + 3 \cdot (-2) = 5 + 1 - 6 = 0$, so $\omega_1 \wedge \omega_1 = \mathbf{0}$ and the test is inconclusive. In fact ω_1 is decomposable: it is $\mathbf{v}_1 \wedge \mathbf{v}_2$ for $\mathbf{v}_1 = (1, 0, 2, 1)$ and $\mathbf{v}_2 = (0, 1, 1, 3)$, by Example (A wedge of two vectors in four-dimensional space).

For ω_2 : $1 \cdot 1 - 0 \$

Exercise. C1: The Plücker relation

Let $\dim V = 4$ with basis $(\mathbf{e}_1, \dots, \mathbf{e}_4)$, let $\text{char } F \neq 2$, and let $\omega = \sum_{i < j} c_{ij} \mathbf{e}_{ij} \in \Lambda^2 V$.

- Prove that $\omega \wedge \omega = 2(c_{12}c_{34} - c_{13}c_{24} + c_{14}c_{23}) \mathbf{e}_{1234}$.
- Deduce that if $c_{12}c_{34} - c_{13}c_{24} + c_{14}c_{23} \neq 0$ then ω is not decomposable.
- Prove the converse: if the expression vanishes, then ω is decomposable. *Hint: if $\omega \neq \mathbf{0}$, some $c_{ij} \neq 0$; after relabeling the basis assume $c_{12} \neq 0$ and look for ω as a wedge of two vectors whose first two coordinates you may normalize.*

Solution.

- By bilinearity, $\omega \wedge \omega = \sum_{S,T} c_S c_T \mathbf{e}_S \wedge \mathbf{e}_T$ over 2-element sets S, T . A term vanishes unless $S \cap T = \emptyset$, so the surviving pairs are $(\{1, 2\}, \{3, 4\})$, $(\{1, 3\}, \{2, 4\})$, $(\{1, 4\}, \{2, 3\})$ and their reverses. By [Theorem 14.10.3](#) with $k = l = 2$, reversing a pair does not change the product, so each contributes twice. Sorting, $\mathbf{e}_{12} \wedge \mathbf{e}_{34} = \mathbf{e}_{1234}$; $\mathbf{e}_{13} \wedge \mathbf{e}_{24} = \mathbf{e}_1 \wedge \mathbf{e}_3 \wedge \mathbf{e}_2 \wedge \mathbf{e}_4 = -\mathbf{e}_{1234}$, one transposition; and $\mathbf{e}_{14} \wedge \mathbf{e}_{23} = \mathbf{e}_1 \wedge \mathbf{e}_4 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 = \mathbf{e}_{1234}$, two transpositions. Collecting gives the stated formula.
- If $\omega = \mathbf{u} \wedge \mathbf{w}$ then $\omega \wedge \omega = \mathbf{u} \wedge \mathbf{w} \wedge \mathbf{u} \wedge \mathbf{w} = \mathbf{0}$, a wedge with a repeated vector. Since $\text{char } F \neq 2$, (a) then forces the bracket to vanish. Contrapositively, a non-zero bracket rules out decomposability.
- If $\omega = \mathbf{0}$ it is decomposable. Otherwise relabel the basis so that $c_{12} \neq 0$; this is legitimate because a permutation of the basis permutes the six coefficients up to sign and changes

Justify your answer.

Solution.

- (a) By the computation of [Exercise \(C1: The Plücker relation\)](#) (a), $\omega \wedge \omega$ is 2 times an element of $\Lambda^4 V$, and $2 = 0$ in $\mathbb{F}_2$. (Equivalently: in $\sum_{S,T} c_{SCT} \mathbf{e}_S \wedge \mathbf{e}_T$ the diagonal terms vanish and the off-diagonal ones pair up, each pair summing to $2c_{SCT} \mathbf{e}_S \wedge \mathbf{e}_T = \mathbf{0}$.)
- (b) Because $\omega \wedge \omega = \mathbf{0}$ is only a **necessary** condition for decomposability, and over $\mathbb{F}_2$ everything satisfies it, so it carries no information at all. The element $\omega = \mathbf{e}_1 \wedge \mathbf{e}_2 + \mathbf{e}_3 \wedge \mathbf{e}_4$ is still not decomposable, by [Theorem 14.10.4](#), whose proof computes $\mathbf{v} \wedge \omega$ for a general $\mathbf{v}$ and never divides by 2.
- (c) No, that one is characteristic-free: [Theorem 14.10.3](#) proves $\omega \wedge \omega = \mathbf{0}$ for odd degree over every field, by pairing off the terms with $S \neq T$ rather than by dividing. Over $\mathbb{F}_2$ the statement is true but vacuous as a test, for the same reason as in (a).

Exercise. C3: The even part of a Clifford algebra

Let $\text{char } F \neq 2$, $\dim V = n \geq 1$, and let (C, ι) be a Clifford algebra of (V, q) , and let $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ be an orthogonal basis of V for β . Let $C^+ \subseteq C$ be the span of the sorted products with k **even**, and C^- the span of those with k odd. Prove that $C = C^+ \oplus C^-$, that C^+ is a subalgebra of C , and that $\dim_F C^+ = \dim_F C^- = 2^{n-1}$. Is C^- a subalgebra?

Hint: what does the sorting procedure of Proposition 13.12.3 do to the parity of the length of a product?

Solution. The sorted products form a basis of C by [Theorem 14.10.8](#), and they are partitioned by the parity of $|S|$, so $C = C^+ \oplus C^-$ and the two dimensions are $\sum_{k \text{ even}} \binom{n}{k}$ and $\sum_{k \text{ odd}} \binom{n}{k$